Part B — Stretch Problem

1. Model Evaluation

```
from sklearn.metrics import accuracy_score, confusion_matrix

y_pred = model.predict(X_test)

print('Accuracy:', accuracy_score(y_test, y_pred))
print('Confusion Matrix:')
print(confusion_matrix(y_test, y_pred))
```

The confusion matrix breaks down predictions into TP, TN, FP, FN — useful for understanding where the model is making mistakes beyond just accuracy.

2. Decision Boundary Plot (Optional)

```
import numpy as np
import matplotlib.pyplot as plt

x_min, x_max = X_train[:, 0].min() - 1, X_train[:, 0].max() + 1
y_min, y_max = X_train[:, 1].min() - 1, X_train[:, 1].max() + 1

xx, yy = np.meshgrid(
    np.arange(x_min, x_max, 0.01),
    np.arange(y_min, y_max, 0.01)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

plt.contourf(xx, yy, Z, alpha=0.4)
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, s=20)
plt.title('Decision Boundary (Train Set)')
plt.show()
```

The decision boundary gives a visual sense of where the model draws the line between buyers and non-buyers in the Age vs Salary space.

3. Comparing Different Test Sizes

```
for test_size in [0.2, 0.25, 0.3]:
    Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=test_size,
random_state=42)
    Xtr = sc.fit_transform(Xtr)
    Xte = sc.transform(Xte)
    m = LogisticRegression().fit(Xtr, ytr)
    acc = accuracy_score(yte, m.predict(Xte))
    print(f'Test size {test_size} -> Accuracy: {acc:.4f}')
```

Accuracy tends to stay stable across splits if the dataset is well-distributed. A big drop with 70/30 might indicate the model is sensitive to how much training data it sees.